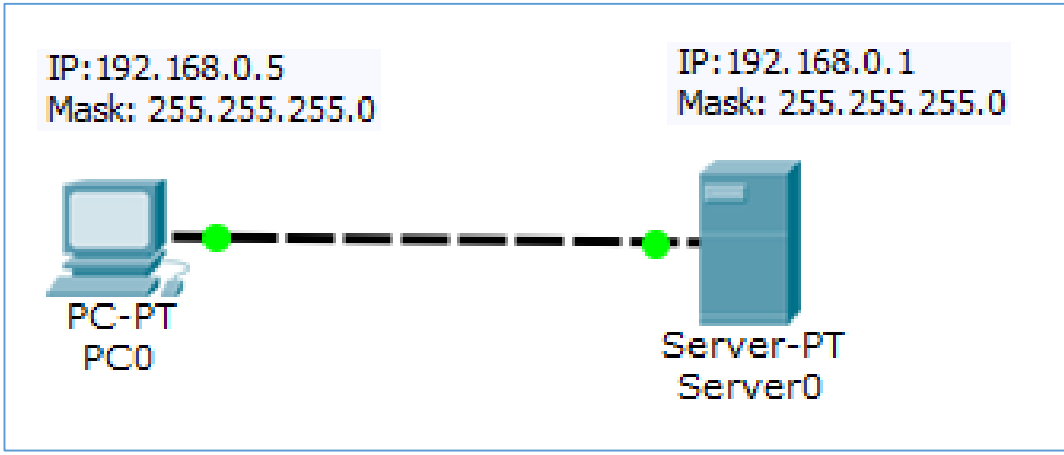
Network’s Lab.

Instructor: Masoud Asghari

Session: 2

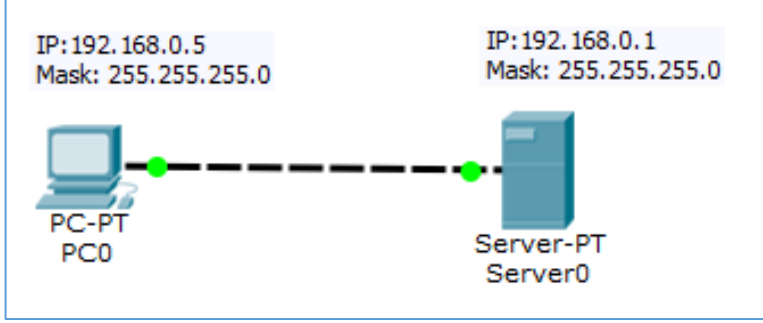
Showcase S1_1,

- Create following topology

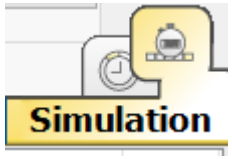


- Set IP and Subnet masks as shown above
- On PC0 Config window, Desktop tab:
  - Open Web browser and type server's IP address
  - Open Command prompt:
    - Ping 192.168.0.1
    - Ping 192.168.0.5
    - Ping 127.0.0.1
    - Ping 192.168.0.6
- On Server0 Config window, Desktop tab:
  - Open Command prompt:
    - Ping 192.168.0.5
    - Ping 192.168.0.6

Showcase S1_1,



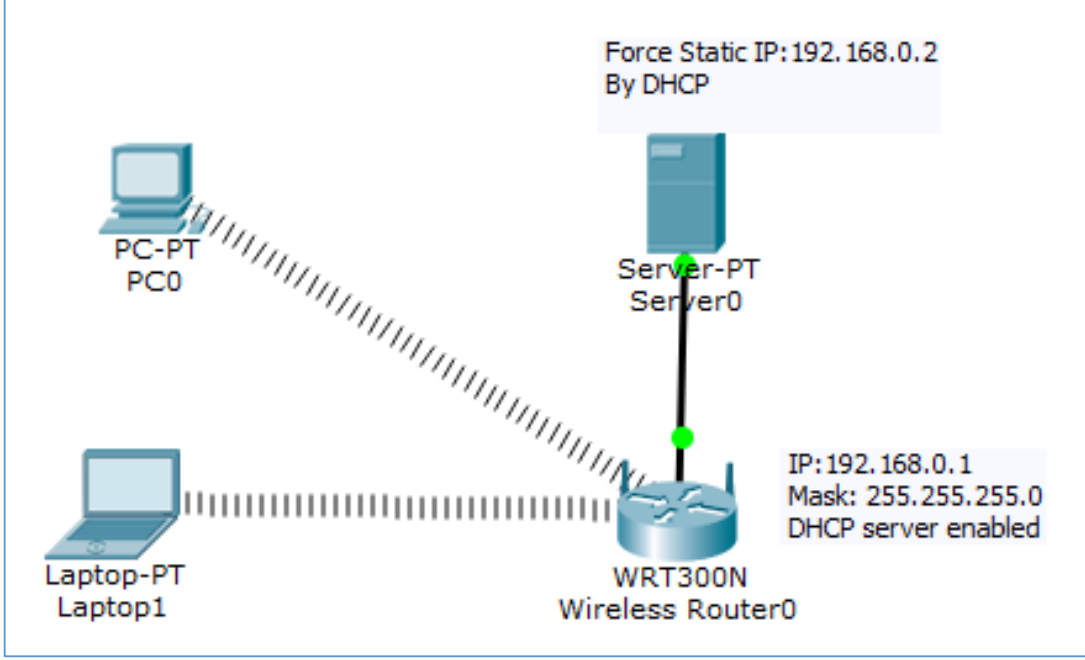
- Use “Add simple PDU” and add an envelope first to PC0 and then to Server0
  - See the status at “PDU list window”
- Use the above technique this time add an envelope first to Server0 and then to PC0
  - See the status at “PDU list window”
- **Change mode to “Simulation”**



  - See how ICMP packets are transmitted (step by step)

Showcase S1_2, *Simple WiFi network*

- Create following topology



- Add **wifi** module to PC0 and Laptop1 by
  - Turn it off, remove Ethernet module, add "**WMP300N**" module, turn it on
- IP Of PC, Server, and Laptop are assigned by DHCP server at the Wireless router
  - Config DHCP of Wireless router to assign static IP "**192.168.0.2**" to Server0
    - Use ***GUI*** of Wireless router and add "***DHCP reservation***" by MAC address of Server0

- Use ping on computers to make sure they are all connected
- Use web browser at Laptop to see "192.168.0.2"

Showcase S1_3, *Simple DNS server*

- **<u>Based on S1_2</u>**
- At Wireless Router0
  - At “GUI” > Static DNS 1: > Put *192.168.0.2*
  - Save setting
- Click Power Cycle Devices And then multiple times on Fast Forward Time
- At Server0
  - At “Services” > “DNS”
  - Add following entries

No.	Name	Type	Detail
0	test.com	A Record	192.168.0.2
1	www.test.com	CNAME	test.com

- At Laptop1
  - Use web browser see “test.com” and “www.test.com”
  - Use Command prompt:
    - nslookup test.com
    - nslookup www.test.com
    - ping www.test.com

In this section

Network related commands available at Windows7 and above

Command1: *ping*

- This utility is used to provide a basic connectivity test between the requesting host and a destination host.
- This is done by using the Internet Control Message Protocol (ICMP) which has the ability to send an echo packet to a destination host and a mechanism to listen for a response from this host.

Command Prompt

```
C:\Users\mas>ping urmia.ac.ir

Pinging urmia.ac.ir [217.219.147.140] with 32 bytes of data:
Reply from 217.219.147.140: bytes=32 time=57ms TTL=50
Reply from 217.219.147.140: bytes=32 time=65ms TTL=50
Reply from 217.219.147.140: bytes=32 time=63ms TTL=50
Reply from 217.219.147.140: bytes=32 time=55ms TTL=50

Ping statistics for 217.219.147.140:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 55ms, Maximum = 65ms, Average = 60ms

C:\Users\mas>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=138ms TTL=37
Reply from 8.8.8.8: bytes=32 time=154ms TTL=37
Reply from 8.8.8.8: bytes=32 time=138ms TTL=37
Reply from 8.8.8.8: bytes=32 time=161ms TTL=37

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 138ms, Maximum = 161ms, Average = 147ms

C:\Users\mas>_
```

Command2: *tracert*

- The tracert utility can used to determine more specific information about the path to the destination host including the route the packet takes and the response time of these intermediate hosts.

```
C:\Users\mas>tracert -d urmia.ac.ir

Tracing route to urmia.ac.ir [217.219.147.140]
over a maximum of 30 hops:

  1     1 ms    <1 ms    <1 ms  192.168.1.1
  2     *        *        *     Request timed out.
  3    58 ms    67 ms    67 ms  10.135.20.241
  4    61 ms    70 ms    63 ms  10.135.16.42
  5    67 ms    65 ms    67 ms  10.21.251.42
  6    72 ms    80 ms    65 ms  10.42.42.129
  7    79 ms    77 ms    73 ms  10.43.44.130
  8    81 ms    85 ms    85 ms  10.21.252.105
  9    76 ms    80 ms    91 ms  217.219.147.190
 10    76 ms    85 ms    81 ms  217.219.147.140

Trace complete.
```

```
C:\Users\mas>tracert 8.8.8.8

Tracing route to google-public-dns-a.google.com [8.8.8.8]
over a maximum of 30 hops:

  1     1 ms     2 ms     1 ms  192.168.1.1
  2     *        *        *     Request timed out.
  3    59 ms    65 ms    67 ms  10.135.20.241
  4    63 ms    65 ms    79 ms  10.135.16.42
  5    65 ms    67 ms    79 ms  10.21.251.42
  6    76 ms    67 ms    69 ms  10.42.42.129
  7    82 ms    66 ms    67 ms  10.43.43.153
  8     *       46 ms    93 ms  10.43.43.69
  9   137 ms   147 ms   147 ms  so-0-2-2.franco71.fra.seabone.net [89.221.34.194]
 10   138 ms   189 ms   146 ms  72.14.205.182
 11   223 ms   221 ms   221 ms  216.239.56.190
 12   135 ms   149 ms   135 ms  216.239.57.188
 13   137 ms   147 ms   145 ms  209.85.251.178
 14   143 ms   161 ms   156 ms  74.125.37.154
 15   157 ms   157 ms   161 ms  209.85.246.135
 16     *        *        *     Request timed out.
 17   132 ms   137 ms   157 ms  google-public-dns-a.google.com [8.8.8.8]

Trace complete.
```

Command3: *ipconfig*

- ipconfig /?
  - Shows help
- ipconfig /all
  - shows the IP configuration information of a host
- ipconfig /flushdns
  - flushes the DNS Resolver Cache
- ipconfig /displaydns
  - Display the contents of the DNS Resolver Cache.
- **An important additional feature of ipconfig is to allow system admins to force refreshing of the DHCP IP address of the host computer to request a different IP address.**
- ipconfig /release
  - is executed to force the client to immediately give up its leased IP from DHCP by sending the server a DHCP release notification which updates the server's status information and marks the old client's IP address as "Available".
- ipconfig /renew
  - is executed to request a new IP address from DHCP server

Command4: *nslookup*

- Query DNS server for dns names (domains)
  - Do not use https:// at the beginning

```
C:\Users\mas>nslookup yahoo.com
Server:  UnKnown
Address:  127.0.0.1

Non-authoritative answer:
Name:    yahoo.com
Addresses:  2001:4998:58:c02::a9
          2001:4998:44:204::a7
          2001:4998:c:a06::2:4008
          98.138.253.109
          206.190.36.45
          98.139.183.24
```

```
C:\Users\mas>nslookup www.yahoo.com
Server:  UnKnown
Address:  127.0.0.1

Non-authoritative answer:
Name:    fd-fp3.wg1.b.yahoo.com
Addresses:  2a00:1288:110:2::4001
          46.228.47.114
          46.228.47.115
Aliases:  www.yahoo.com
```

```
C:\Users\mas>nslookup yahoo.com 8.8.8.8
Server:  google-public-dns-a.google.com
Address:  8.8.8.8

Non-authoritative answer:
Name:    yahoo.com
Addresses:  2001:4998:58:c02::a9
          2001:4998:c:a06::2:4008
          2001:4998:44:204::a7
          98.139.183.24
          206.190.36.45
          98.138.253.109
```

Command5: *netstat*

- **netstat (network statistics)**
  - is a command-line tool that displays network connections for the TCP, routing tables, and a number of network interface and network protocol statistics.
- netstat -a
  - Displays all connections and listening ports.
- netstat -e
  - Displays Ethernet statistics.

Command 6: *Route*

- This utility is used to display the current status of the routing table on a host.
  - You can add or delete new routes
- *route print*

Command 7: *arp*

- Shows *arp* table, add and delete entries

- arp –a
  - Displays current ARP entries

Other commands

You may search for them

- **Pathping**
- **Netsh**
  - Network Command Shell
- **Getmac**
- **Hostname**
- **Net**
- **Nbtstat**
  - TCP/IP NetBIOS Information
  - Displays statistics for the NetBIOS over TCP/IP (NetBT) protocol, NetBIOS name tables for both the local computer and remote computers, and the NetBIOS name cache.

Home work

1.1. At wireless router at “GUI” >
Wireless >
Advanced Wireless setting

- Find what do these do

Beacon Interval:	100	(Default: 100, Milliseconds, Range: 1-65535)
DTIM Interval:	1	(Default: 1, Range: 1-255)
Fragmentation Threshold:	2346	(Default: 2346, Range: 256-2346)
RTS Threshold:	2347	(Default: 2347, Range: 0-2347)